

Table 1: Simulation results under different forms of heteroskedasticity

	Results without rounding error				Results with rounding error			
	β_1		β_2		β_1		β_2	
Estimator:	Bias	S.E.	Bias	S.E.	Bias	S.E.	Bias	S.E.
Case 1:	$V[y_i x] = 1$							
PPML	-0.00004	0.016	0.00009	0.027	0.01886	0.017	0.02032	0.029
NLS	-0.00006	0.008	-0.00003	0.017	0.00195	0.008	0.00274	0.018
GPML	0.01276	0.068	0.00754	0.082	0.10946	0.096	0.09338	0.108
OLS	0.39008	0.039	0.35568	0.054	—	—	—	—
ET-Tobit	-0.47855	0.030	-0.47786	0.032	-0.49981	0.030	-0.49968	0.032
OLS ($y > 0.5$)	-0.16402	0.027	-0.15487	0.038	-0.22121	0.026	-0.21339	0.036
OLS ($y + 1$)	-0.40237	0.014	-0.37683	0.022	-0.37752	0.015	-0.34997	0.024
Case 2:	$V[y_i x] = \mu(x_i\beta)$							
PPML	-0.00011	0.019	0.00009	0.039	0.02190	0.020	0.02334	0.041
NLS	0.00046	0.033	0.00066	0.057	0.00262	0.033	0.00360	0.057
GPML	0.00376	0.043	0.00211	0.062	0.13243	0.073	0.11331	0.087
OLS	0.21076	0.030	0.19960	0.049	—	—	—	—
ET-Tobit	-0.42394	0.028	-0.42316	0.033	-0.45518	0.028	-0.45513	0.033
OLS ($y > 0.5$)	-0.17868	0.026	-0.17220	0.043	-0.24405	0.026	-0.23889	0.040
OLS ($y + 1$)	-0.42371	0.015	-0.39931	0.025	-0.39401	0.016	-0.36806	0.028
Case 3:	$V[y_i x] = \mu(x_i\beta)^2$							
PPML	-0.00526	0.091	-0.00228	0.130	0.02332	0.091	0.02812	0.133
NLS	0.23539	3.066	0.07323	1.521	0.23959	3.082	0.07852	1.521
GPML	-0.00047	0.041	-0.00029	0.083	0.17134	0.068	0.14442	0.104
OLS	0.00015	0.032	-0.00003	0.064	—	—	—	—
ET-Tobit	-0.31908	0.044	-0.32161	0.058	-0.36480	0.043	-0.36789	0.056
OLS ($y > 0.5$)	-0.34480	0.039	-0.34614	0.064	-0.41006	0.037	-0.41200	0.060
OLS ($y + 1$)	-0.51804	0.021	-0.50000	0.038	-0.48564	0.022	-0.46597	0.040
Case 4:	$V[y_i x] = \mu(x_i\beta) + \exp(x_{2i})\mu(x_i\beta)^2$							
PPML	-0.00696	0.103	-0.00647	0.144	0.02027	0.104	0.01856	0.146
NLS	0.35139	7.516	0.08801	1.827	0.35672	7.521	0.09239	1.829
GPML	0.00322	0.057	-0.00137	0.083	0.12831	0.085	0.10245	0.129
OLS	0.13270	0.039	-0.12542	0.075	—	—	—	—
ET-Tobit	-0.29908	0.049	-0.42731	0.063	-0.34351	0.047	-0.46225	0.060
OLS ($y > 0.5$)	-0.41391	0.042	-0.41391	0.070	-0.45188	0.040	-0.46173	0.066
OLS ($y + 1$)	-0.51440	0.021	-0.58087	0.041	-0.48627	0.022	-0.56039	0.044

In the first set of experiments, the results of the gamma PML estimator are very good. Indeed, when no measurement error is present, the biases and standard errors of the GPML estimator are always among the lowest. However, this estimator is very sensitive to the form of measurement error considered in the second set of experiments, consistently leading to sizable biases. These results, like those of the NLS, clearly illustrate the danger of using a PML estimator that gives extra weight to the noisier observations.

As for the performance of the Poisson PML estimator, the results are very encouraging. In fact, when no rounding error is present, its performance is reasonably good in all cases. Moreover, although some loss of efficiency is noticeable as one moves away from Case 2, in which it is an optimal estimator, the biases of the PPML are always small.¹⁹ Moreover, the results obtained with rounded data suggest that the Poisson based PML estimator is relatively robust to this form of measurement error of the dependent variable. Indeed, the bias introduced by the rounding-off errors in the dependent variable is relatively small and, in some cases, it even compensates the bias found in the first set of experiments. Therefore, because it is simple to implement and reliable under a wide variety of situations, the Poisson PML estimator has the essential characteristics needed to make it the new work-horse for the estimation of constant-elasticity models.

Obviously, the sign and magnitude of the bias of the estimators studied here depend on the particular specification considered. Therefore, the results of these experiments cannot serve as an indicator of what can be expected in other situations. However, it is clear that, apart from the Poisson PML method, all estimators will often be very misleading.

These experiments were also used to study the finite sample performance of the Gauss-Newton regression (GNR) test for the adequacy of the Poisson PML based on (13) and of the Park test advocated by Manning and Mullahy (2001), which, as explained above, is only valid to check for the adequacy of the estimator based on the log-linear model.²⁰ Given that the Poisson PML estimator is the only estimator with a reasonable behavior under all the cases considered, these tests were performed using residuals and estimates of $\mu(x_i\beta)$ from the Poisson regression. Table 2 contains the rejection frequencies of the null hypothesis at the 5 percent nominal level for both tests in the four cases considered in the two sets of experiments. In this table the rejection frequencies under the null hypothesis are given in bold.

¹⁹These results are in line with those reported by Manning and Mullahy (2001).

²⁰To illustrate the pitfalls of the procedure suggested by Manning and Mullahy (2001) we note that the means of the estimates of λ_1 obtained using (11) in cases 1, 2 and 3 (without measurement error) were 0.58955, 1.29821 and 1.98705, whereas the true values of λ_1 in these cases are, respectively, 0, 1 and 2.

Since both tests have adequate behavior under the null and reveal reasonable power against a wide range of alternatives, the results suggest that these tests are important tools to assess the adequacy of the standard OLS estimator of the log-linear model and of the proposed Poisson PML estimator.

Table 2: Rejection frequencies at the 5% level for the two specification tests

Without Measurement Error				
Test	Case 1	Case 2	Case 3	Case 4
GNR	0.91980	0.05430	0.58110	0.49100
Park	1.00000	1.00000	0.06680	0.40810
With Measurement Error				
Test	Case 1	Case 2	Case 3	Case 4
GNR	0.91740	0.14980	0.57170	0.47580
Park	1.00000	1.00000	1.00000	1.00000

5. The gravity equation

In this section, we use the PPML estimator to quantitatively assess the determinants of bilateral trade flows, uncovering significant differences in the roles of various measures of size and distance from those predicted by the logarithmic tradition. We perform the comparison of the two techniques using both the traditional and the Anderson-van Wincoop (2003) specifications of the gravity equation.

For the sake of completeness, we also compare the PPML estimates with those obtained from alternative ways researchers have used to deal with zero-values for trade. In particular, we present the results obtained with the Tobit estimator used in Eaton and Tamura (1994), OLS estimator applied to $\ln(1 + T_{ij})$, and a standard non-linear least squares estimator. The results obtained with these estimators are presented, as before, for both the traditional and the Anderson-van Wincoop specifications.

5.1 The data

The analysis covers a cross section of 136 countries in 1990. Hence, our data set consists of 18,360 observations of bilateral import flows (136×135 country pairs). The list of countries is reported in Table A1 in the Appendix. Information on bilateral trade comes

from Feenstra et al. (1997). Data on real GDP per capita and population come from the World Bank’s *World Development Indicators* (2002). Data on location and dummies indicating contiguity, common language (official and second languages), colonial ties (direct and indirect links), and access to water are constructed from the CIA’s *World Factbook*. The data on language and colonial links are presented on Tables A2 and A3 in the appendix.²¹ Bilateral distance is computed using the great circle distance algorithm provided by Andrew Gray (2001). Remoteness – or relative distance – is calculated as the (log of) GDP-weighted average distance to all other countries (see Wei, 1996). Finally, information on preferential-trade agreements comes from Frankel (1997), complemented with data from the World Trade Organization. The list of preferential trade agreements (and stronger forms of trade agreements) considered in the analysis is displayed in Table A4 in the Appendix. Table A5 in the Appendix provides a description of the variables and displays the summary statistics.

5.2 Results

5.2.1 The Traditional Gravity Equation

Table 3 presents the estimation outcomes resulting from the various techniques for the traditional gravity equation. The first column reports OLS estimates using the logarithm of trade as the dependent variable; as noted before, this regression leaves out pairs of countries with zero bilateral trade (only 9,613 country pairs, or 52 percent of the sample, exhibit positive import flows).

The second column reports the OLS estimates using $\ln(1 + T_{ij})$ as dependent variable, as a way of dealing with zeroes. The third column presents Tobit estimates based on Eaton and Tamura (1994). The fourth column shows the results of standard NLS. The fifth column reports Poisson estimates using only the subsample of positive-trade pairs.

²¹ Alternative estimates based on Boisso and Ferrantino (1997)’s index of language similarity are available, at request, from the authors.